

SUPPORTLYTICS

IT Support Analytics & Performance Intelligence

A comprehensive analytical evaluation of enterprise IT service desk performance, multi-channel support ticket lifecycles, semantic issue clustering, SLA compliance adherence, and regional operational efficiency. Engineered across Milestones 1–4 and Modules 1–8 with Python data pipelines, advanced DAX measures, and deployed through an interactive 6-page Microsoft Power BI dashboard.

Project Attribute	Enterprise Specification & Deliverable Details
Project Classification	Internship Project — IT Support Analytics
Milestone & Module Scope	Milestones 1–4 (Weeks 1–8) Modules 1–8 (Complete Lifecycle)
Primary Deliverable	Microsoft Power BI Interactive Dashboard (Supportlytics_Dashboard.pbix)
Analytical Scope	10,000 Support Tickets 38 Schema Attributes 3 Core Clusters 3 Support Tiers
Key Verified Baseline KPIs	Avg Turnaround: 61.1032 hrs SLA Met: 50.39% Similarity Index: 0.7446
Technology Stack	Microsoft Power BI Desktop, DAX, Python 3.10+, Pandas, NumPy, Jupyter
Authoritative Data Source	Supportlytics_PowerBI_Data.xlsx (Sheet: Data)
Documentation Version & Date	Version 1.0 Final Enterprise Release August 2026

PAGE 02 | EXECUTIVE SUMMARY

Strategic Overview of Supportlytics Intelligence System

1. Strategic Project Context & Operational Mandate

Supportlytics is an enterprise-grade analytics and business intelligence solution developed as an Internship Project in IT Support Analytics. Modern enterprise IT support departments handle thousands of service requests, incident reports, and systemic technical problems monthly across distributed global regions. Without centralized intelligence, IT leadership faces operational bottlenecks, SLA breach liabilities, queue imbalances, and repetitive issue overhead. This project establishes an automated analytics pipeline across Milestones 1–4 culminating in a production-ready Microsoft Power BI dashboard (Supportlytics_Dashboard.pbix).

2. Verified Core Key Performance Indicators (KPIs)

KPI Metric	Verified Value	Calculation & Business Definition	Operational Target Status
Total Ticket Volume	10,000 Tickets	COUNTROWS('Data') across multi-channel intakes	Baseline Demand Assessed
Avg Resolution Time	61.1032 Hours	AVERAGE('Data'[Resolution_Hours]) creation to close	Operational Baseline Benchmark
Cluster Similarity Index	0.7446 (74.46%)	AVERAGE('Data'[Similarity_Score]) across clusters	High Semantic Cohesion (> 0.70)
SLA Compliance Rate	50.39% (5,039)	DIVIDE(SLA Met [5,039], Total Tickets [10,000])	5,039 Met / 4,961 Breached
Avg Customer Satisfaction	2.9818 / 5.00	AVERAGE('Data'[Customer_Satisfaction]) post-close	Baseline Service Quality Level
High/Critical Unresolved	2,504 Tickets	CALCULATE(Unresolved, Priority IN {High, Critical})	Active Pending Backlog Focus

3. Major Verified Findings & Business Value

- 1. Turnaround Metric: Average resolution time stands at 61.1032 hours across all tiers, establishing a consistent operational baseline.
- 2. Contractual SLA Distribution: With a 50.39% SLA compliance rate (5,039 Met tickets), 4,961 tickets exceeded contracted windows, highlighting queue triage needs.
- 3. Issue Clustering: Three verified issue clusters (Login, Network, Software) exhibit a 0.7446 similarity index, highlighting clear self-service deflection potential.

[EXECUTIVE FOCUS] Centralized dashboard monitoring and prioritized management of the 2,504 High/Critical unresolved tickets provide direct operational visibility for service desk leads.

PAGE 03 | PROJECT OVERVIEW

Domain Architecture, Scope Boundaries & Target Users

1. Background and Domain Context

In the modern enterprise, the IT Service Desk functions as the operational backbone ensuring continuous availability, hardware reliability, network connectivity, access security, and software support for global personnel. As organizations scale, multi-channel intakes (Email, Web Portal, Live Chat, Phone) create voluminous ticket streams. Without unified business intelligence, support teams operate reactively—treating incidents in isolation rather than leveraging macro-level behavioral patterns to optimize team allocation, identify underlying defects, and prevent recurring breaches.

2. Analytical Methodology: Workflow Sequence

Supportlytics follows a structured analytical pipeline: Data Ingestion → Cleansing & Validation → Feature Preparation → Exploratory & Cluster Analytics → Performance & SLA Intelligence → Power BI Modeling → Executive Decision Support.

3. Target Stakeholders & Operational Value

Stakeholder Role	Primary Operational Responsibility	Supportlytics Decision Support Value
IT Operations Director	Overall ITSM governance, resource allocation, SLA adherence	Macro KPI visibility, regional benchmarking, breach liability tracking
Service Desk Manager	Daily queue balancing, escalation management, team productivity	Workload distribution, priority monitoring, high-urgency backlog triage
Tier 1/2 Team Leads	Incident resolution speed, first-contact resolution, agent guidance	Turnaround benchmarks across L1, L2, Security teams, category friction points
Continuous Improvement Lead	Problem management, recurring issue reduction, deflection	Cluster analysis (Login, Network, Software), recurring pattern identification

4. Primary Project Deliverables

- Cleaned Dataset: Standardized 10,000-ticket dataset with 38 attributes in Supportlytics_PowerBI_Data.xlsx.
- Power BI Dashboard: 6-page interactive report (Supportlytics_Dashboard.pbix) with dynamic cross-filtering.
- DAX KPI Architecture: 11 Primary KPI measures + 6 Supporting measures in a dedicated calculation engine.

PAGE 04 | BUSINESS PROBLEM

Operational Bottlenecks & Service Delivery Vulnerabilities

1. The Enterprise IT Support Challenge

IT Service Desks manage heavy ticket traffic to minimize downtime, expedite incident resolution, and uphold contracted Service Level Agreements (SLAs). However, operational fragmentation often obscures the root causes of service delays. Analysis of the enterprise support dataset highlights five core operational challenges:

2. Problem-to-Impact Operational Matrix

Vulnerability Area	Observed Empirical Evidence	Operational Risk & Business Consequence
Extended Resolution Duration	Average turnaround duration of 61.1032 hours across 10,000 tickets	Prolonged user waiting time, delayed operational workflows
SLA Non-Compliance Ratio	50.39% SLA met rate (4,961 out of 10,000 tickets breached)	Potential contractual penalties, lower customer satisfaction ratings (2.9818)
High-Priority Backlog Volume	2,504 High and Critical unresolved tickets in active queue	Elevated operational risk from pending high-urgency system issues
Repetitive Incident Overhead	Similarity score averaging 0.7446 across Login, Network, Software clusters	Manual engineering effort spent handling recurring access and software tickets
Geographic Workload Differences	Variations across North America, EMEA, and APAC regions	Resource allocation differences across geographic support queues

3. Static Reporting Deficiencies vs. Interactive BI Requirements

Traditional service desks often rely on periodic static summaries or spreadsheet trackers. These mechanisms provide retrospective views rather than interactive diagnostic capabilities. Static reports cannot dynamically cross-filter resolution times across categories, isolate country-specific patterns, or track unresolved high-priority queues in real time.

[CENTRALIZED ANALYTICS IMPERATIVE] An interactive business intelligence dashboard enables support leadership to evaluate workload distribution and queue health dynamically across categories, teams, and geographic locations.

PAGE 05 | PROJECT OBJECTIVES

Milestones 1–4 & Modules 1–8 Roadmap Architecture

1. Primary Project Objective

The overarching goal of the Supportlytics project is to engineer an end-to-end data analytics and business intelligence system that converts 10,000 raw IT support ticket records into actionable operational insights, empowering service desk leaders to systematically monitor resolution duration, evaluate SLA compliance, identify recurring issue patterns, and optimize queue management.

2. Milestones 1–4 & Modules 1–8 Roadmap

Milestone	Timeline	Covered Modules	Key Objectives & Deliverables
Milestone 1	Weeks 1–2	Module 1: Setup & Initialization Module 2: Ingestion & Schema	Setup workspace, verify raw 10,000 records, inspect 38 attributes, establish primary key uniqueness.
Milestone 2	Weeks 3–4	Module 3: Cleaning & Features Module 4: EDA & Clustering	Impute nulls, derive Resolution_Hours and Priority_Score, profile types/categories, evaluate 3 clusters.
Milestone 3	Weeks 5–6	Module 5: Performance & SLA Module 6: Geographic Analysis	Benchmark turnaround across L1/L2/Security tiers, compute 50.39% SLA, map NA/EMEA/APAC regional density.
Milestone 4	Weeks 7–8	Module 7: Power BI Dashboard Module 8: Docs & Repository	Author 6-page interactive report (Supportlytics_Dashboard.pbix), build 11 DAX KPIs, compile final 30-page documentation.

3. Operational Alignment & Measurable Outcomes

Each technical objective directly supports key service desk operational priorities: workload monitoring, queue transparency, turnaround tracking, and SLA visibility. The resulting analytical artifacts provide a permanent, reusable decision-support framework.

[PROJECT DELIVERY RIGOR] The structured 8-week execution plan guarantees that raw transaction records are systematically cleansed, enriched, modeled, and visualized for maximum executive utility.

PAGE 06 | PROJECT SCOPE & WORKFLOW

Ten-Stage Deterministic Analytics Pipeline Architecture

1. Project Scope Boundaries

- In-Scope: Processing the authoritative 10,000-ticket dataset, 38 schema columns, exploratory data analysis, feature engineering, cluster and similarity analysis, DAX measure modeling, Star-Schema relationship design, Power BI report authoring, and analytical recommendations.
- Out-of-Scope: Direct live API streaming from external ticketing systems and production deployment of automated bot agents.

2. Ten-Stage Analytical Pipeline Architecture

Pipeline Stage	Core Technical Activities & Transformation Logic
1. Raw Data Ingestion	Acquisition of authoritative enterprise support log in Excel/CSV format (10,000 records).
2. Data Validation & Cleansing	Schema verification, missing value check, type casting, Ticket_ID uniqueness validation.
3. Feature Engineering	Derivation of Resolution_Hours, Priority_Score, Created_Year, Created_Month, Created_Month_Number.
4. Exploratory Data Analysis	Statistical profiling of ticket types, categories, intake channels, departments, and priority ratios.
5. Cluster & Similarity Analysis	Similarity score evaluation across Login, Network, and Software issue clusters.
6. Performance & Turnaround	Turnaround duration computation across support teams (L1, L2, Security) and priority levels.
7. SLA & CSAT Modeling	SLA compliance calculation (50.39% Met) and customer satisfaction score aggregation.
8. Geographic & Spatial Analysis	Regional workload mapping across North America, EMEA, APAC, and 5 country locations.
9. Power BI Data Modeling	Creation of star schema (Data fact table + Calendar dimension) and 11 DAX KPI measures.
10. Dashboard & Documentation	Authoring of 6 specialized dashboard pages, validation, and final 30-page documentation compilation.

[PIPELINE CONSISTENCY] The analytical workflow ensures consistency: metrics in the final Power BI dashboard reconcile directly with the underlying dataset and DAX calculations.

PAGE 07 | DATASET OVERVIEW

Authoritative Source Structure & Field Group Taxonomy

1. Authoritative Data Source Specification

The definitive data source for this project is Supportlytics_PowerBI_Data.xlsx (Sheet: Data), containing exactly 10,000 records and 38 attributes. Every record represents a distinct IT support ticket logged within the enterprise ecosystem. A 100% unique primary key check confirms exactly 10,000 distinct Ticket_ID values with zero nulls or duplicates.

2. Eight Logical Schema Field Groups

Field Group	Included Dataset Attributes	Operational & Analytical Role
1. Ticket Lifecycle	Ticket_ID, Created_Date, Resolution_Date, First_Response_Date, Status, Resolution_Code, Resolution_Notes	Tracks ticket progression, turnaround duration, lifecycle milestones, and final closure codes.
2. Classification	Priority, Priority_Score, Ticket_Type, Category, Product, Tags, Description	Enables workload classification across incidents, problems, requests, and technical products.
3. Assignment	Assigned_Agent, Team, Department, Channel, Escalation_Level	Defines operational ownership, tier escalation paths, departmental demand, and intake channels.
4. Service Outcomes	SLA_Status, Reopened, Customer_Satisfaction, Resolution_Duration, Resolution_Hours	Quantifies service delivery quality, SLA compliance, customer sentiment, and duration in hours.
5. Similarity & Clusters	Cluster_ID, Cluster_Name, Similarity_Score, Similarity_Level	Captures semantic groupings (Login, Network, Software) and similarity scores.
6. Geography	Country, Region, Latitude, Longitude	Supports regional workload aggregation (NA, EMEA, APAC) and geospatial coordinate mapping.
7. Technical Context	Device_Type, Operating_System	Provides hardware and platform context (Windows, macOS, Linux, Mobile, Laptop, Desktop).
8. Date Helpers	Created_Year, Created_Month, Created_Month_Number	Facilitates chronological trend analysis, quarterly rollups, and proper calendar month sorting.

PAGE 08 | DATA DICTIONARY

Comprehensive Attribute Definitions & Technical Specifications

Field Name	Data Type	Sample Values	Business Description & Operational Utility
Ticket_ID	String (PK)	TICK-000001, TICK-010000	Unique primary identifier for each support ticket record.
Created_Date	DateTime	2024-01-01 08:30:00	Timestamp when ticket was initially submitted and logged in system.
Resolution_Date	DateTime	2024-01-03 21:36:12	Timestamp when ticket was marked resolved by support agent.
First_Response_Date	DateTime	2024-01-01 09:15:00	Timestamp of first recorded agent response to ticket.
Status	String	Open, In Progress, Resolved, Closed	Current lifecycle state of the support ticket.
Priority	String	Critical, High, Medium, Low	Urgency level assigned to ticket based on business impact.
Ticket_Type	String	Incident, Problem, Request	ITIL ticket classification defining nature of support requirement.
Category	String	Security, Hardware, Software, Email, Network	Primary technical domain responsible for addressing the issue.
Product	String	Enterprise Portal, VPN, Office 365	Specific software product or infrastructure system involved.
Assigned_Agent	String	Agent_101, Agent_142...	Identifier of support engineer assigned to ticket.
Team	String	L1, L2, Security	Assigned support tier responsible for ticket remediation.
Department	String	IT, HR, Finance, Operations, Sales	Internal business unit submitting the ticket.
Channel	String	Email, Portal, Chat, Phone	Customer intake channel through which ticket was submitted.
Resolution_Hours	Decimal	4.25, 61.10, 142.50	Elapsed duration between Created_Date and Resolution_Date in hours.
SLA_Status	String	Met, Breached	Compliance outcome against contracted resolution timeframe.
Customer_Satisfaction	Decimal	1.0, 2.0, 3.0, 4.0, 5.0	Post-resolution CSAT feedback rating provided by user (1–5 scale).
Cluster_Name	String	Login, Network, Software	Semantic grouping capturing primary issue archetype.
Similarity_Score	Decimal	0.0000 to 1.0000	Similarity metric evaluating proximity to cluster profile.
Country	String	India, Canada, USA, UK, Germany	Geographic country location of the user submitting the ticket.
Region	String	North America, EMEA, APAC	Macro geographical theater for regional reporting.
Latitude / Longitude	Decimal	37.7749, -122.4194...	Geospatial coordinates for map visualization.

PAGE 09 | DATA PREPARATION & VALIDATION

Ten-Step Cleaning Lifecycle & Integrity Reconciliation

1. Ten-Step Data Engineering Lifecycle

To guarantee analytical rigor, the dataset underwent a structured 10-step preparation pipeline: (1) Source Ingestion, (2) Schema & Dimension Inspection (10,000 rows × 38 cols), (3) Primary Key Uniqueness Audit, (4) Null Value & Whitespace Sanitization, (5) ISO Timestamp Parsing, (6) Derived Feature Engineering (Resolution_Hours = Duration in hours), (7) Priority Score Encoding (1-4), (8) Date Dimension Key Extraction, (9) Geospatial Coordinate Verification, and (10) Power BI KPI Reconciliation against dataset baselines.

2. Data Quality & Integrity Validation Audit

Validation Check	Integrity Rule & Criteria	Validation Test Result	Audit Status
Primary Key Uniqueness	COUNT(DISTINCT Ticket_ID) == 10,000	10,000 unique keys; 0 duplicates	PASSED (100%)
Null Key Audit	Zero nulls across core operational columns	0 nulls in Status, Priority, Type, Team	PASSED (100%)
Duration Integrity	Resolution_Date >= Created_Date	All durations non-negative; Mean: 61.1032 hrs	PASSED (100%)
SLA Status Consistency	SLA_Status IN {'Met', 'Breached'}	Met: 5,039 (50.39%), Breached: 4,961 (49.61%)	PASSED (100%)
Cluster Integrity	Cluster_Name IN {'Login', 'Network', 'Software'}	Network: 3,370, Software: 3,346, Login: 3,284	PASSED (100%)
Team Integrity	Team IN {'L1', 'L2', 'Security'}	L1: 3,365, L2: 3,358, Security: 3,277	PASSED (100%)

3. Raw Dataset vs. Power BI-Ready Dataset Distinction

The raw dataset (IT_Support_Tickets.csv) contains raw transactional logs. The Power BI-ready dataset (Supportlytics_PowerBI_Data.xlsx) integrates all engineered duration metrics, date helpers, cluster attributes, and verified coordinates.

[QUALITY CERTIFICATION] All 10,000 records satisfy data quality standards. No records were discarded, maintaining complete data fidelity across analytical tools.

PAGE 10 | POWER BI DATA MODEL

Star-Schema Architecture & Relational Structure

1. Relational Architecture & Star Schema Design

The Power BI analytical engine relies on a Star-Schema data model designed for responsive query performance, cross-filtering, and DAX measure evaluation. The architecture connects the central fact table containing granular ticket records with a dedicated Calendar date dimension.

2. Model Entities & Relational Specifications

Table Name	Table Role	Key Schema Attributes	Cardinality & Filtering
Data	Fact & Dimension Core	10,000 rows, 38 columns (Ticket_ID, Created_Date, Priority, Status, Team, Resolution_Hours, Similarity_Score)	Target table; Many-side (*); Receives filter propagation from Calendar.
Calendar	Date Dimension	Continuous date sequence (Date, Year, Month Number, Month Name, Quarter, Year-Month)	Source table; One-side (1); Propagates single-direction filters to Data[Created_Date].

3. Relational Mechanics & Filter Propagation

- Relationship: Calendar[Date] (1) —[Active, Single-Direction]—> Data[Created_Date] (*)
- Active State: Ensures that slicers on Year, Quarter, or Month automatically filter visual containers across the report.
- Single-Direction Filtering: Maintains predictable filter flow from the date dimension to the transactional fact table.
- Schema Simplicity: Avoids complex snowflake nesting, ensuring fast report rendering across all dashboard pages.

[ENGINEERING BEST PRACTICE] Star-schema modeling minimizes memory consumption and enables sub-second visual cross-filtering across all 6 report pages.

PAGE 11 | CALENDAR TABLE & RELATIONSHIPS

Date Dimension Architecture & Chronological Sorting

1. Dedicated Date Dimension Architecture

A dedicated date dimension table is essential in enterprise Power BI models to support time intelligence calculations, consistent date filtering, and continuous timeline slicing regardless of gaps in transactional data.

2. Calendar Table Schema & Calculated Columns

Column Name	DAX / Power Query Expression	Business Role & Operational Utility
Date	CALENDAR(MIN('Data'[Created_Date]), MAX('Data'[Created_Date]))	Primary key of date dimension; unique daily sequence.
Year	YEAR([Date])	Extracts 4-digit calendar year for annual aggregations.
Month Number	MONTH([Date])	Numeric month (1–12) used as 'Sort by Column' for Month Name.
Month Name	FORMAT([Date], "mmmm")	Full text month name (January, February...) for chart axes.
Quarter	"Q" & FORMAT([Date], "q")	Calendar quarter designation (Q1, Q2, Q3, Q4) for quarterly trends.
Year-Month	FORMAT([Date], "yyyy-mm")	Standardized chronological year-month label for longitudinal views.

3. Chronological Sorting & Relationship Architecture

- Month Sorting Rule: Power BI defaults to alphabetical month sorting (April, August, December...). Setting 'Sort by Column' on Month Name pointing to Month Number ensures proper chronological ordering (January, February, March...).
- Filter Flow: Single-direction 1-to-many relationship propagates filters seamlessly from the Calendar table to the fact table without ambiguous circular paths.

[TIME INTELLIGENCE DESIGN] Proper date dimension modeling ensures flawless time-intelligence filtering across all 6 Power BI dashboard pages.

PAGE 12 | DAX MEASURES & KPI FRAMEWORK

Eleven Primary KPI Measures & Supporting Metric Library

1. Eleven Required Primary KPI Measures

The Power BI dashboard relies on exactly 11 primary KPI measures evaluated dynamically within the active filter context:

Primary KPI Measure	Exact DAX Calculation Formula	Verified Baseline Value & Role
1. Total Tickets	COUNTROWS('Data')	10,000 Tickets (Total operational volume)
2. Average Resolution Time (Hours)	AVERAGE('Data'[Resolution_Hours])	61.1032 Hours (Mean turnaround time)
3. Cluster Similarity Index	AVERAGE('Data'[Similarity_Score])	0.7446 (Mean issue similarity score)
4. Performance Score	AVERAGE('Data'[Customer_Satisfaction]) * 20	59.64 / 100 (Overall service score)
5. SLA Compliance %	DIVIDE(CALCULATE(COUNTROWS('Data'), 'Data'[SLA_Status] = "Met"), [Total Tickets], 0)	50.39% (Formatted percentage)
6. SLA Compliance Rate	DIVIDE(CALCULATE(COUNTROWS('Data'), 'Data'[SLA_Status] = "Met"), [Total Tickets], 0)	0.5039 (Decimal compliance rate)
7. Most Frequent Category	VAR T = TOPN(1, SUMMARIZE('Data', 'Data'[Category], "C", [Total Tickets]), [C], DESC) RETURN CONCATENATEX(T, 'Data'[Category], ",")	Security (2,068 tickets)
8. Top Performing Region	VAR R = TOPN(1, FILTER(ADDCOLUMNS(VALUES('Data'[Region]), avg) "A", [Average Resolution Time (Hours)]), NOT ISBLANK([A])), [A], ASC) RETURN CONCATENATEX(R, 'Data'[Region], ",")	North America (60.6909 hrs avg)
9. Total Unresolved Tickets	CALCULATE([Total Tickets], 'Data'[Status] <> "Resolved", 'Data'[Status] <> "Closed")	4,984 Tickets (Open + In Progress)
10. High Priority Unresolved Tickets	CALCULATE([Total Tickets], 'Data'[Priority] IN {"Critical", "High"}, 'Data'[Status] <> "Resolved", 'Data'[Status] <> "Closed")	2,504 Tickets (Critical + High pending)
11. Average Customer Satisfaction	AVERAGE('Data'[Customer_Satisfaction])	2.9818 / 5.00 (Customer rating)

2. Supporting Operational Measures

• Resolved Tickets (2,518) | Open Tickets (2,470) | Closed Tickets (2,498) | SLA Met (5,039) | SLA Breached (4,961).

PAGE 13 | EXPLORATORY DATA ANALYSIS

Module 4 Foundation Workload Profiling & Distributions

1. Operational Workload Distributions

Exploratory profiling of the 10,000 support records reveals balanced operational demand across ITIL ticket types and technical support domains:

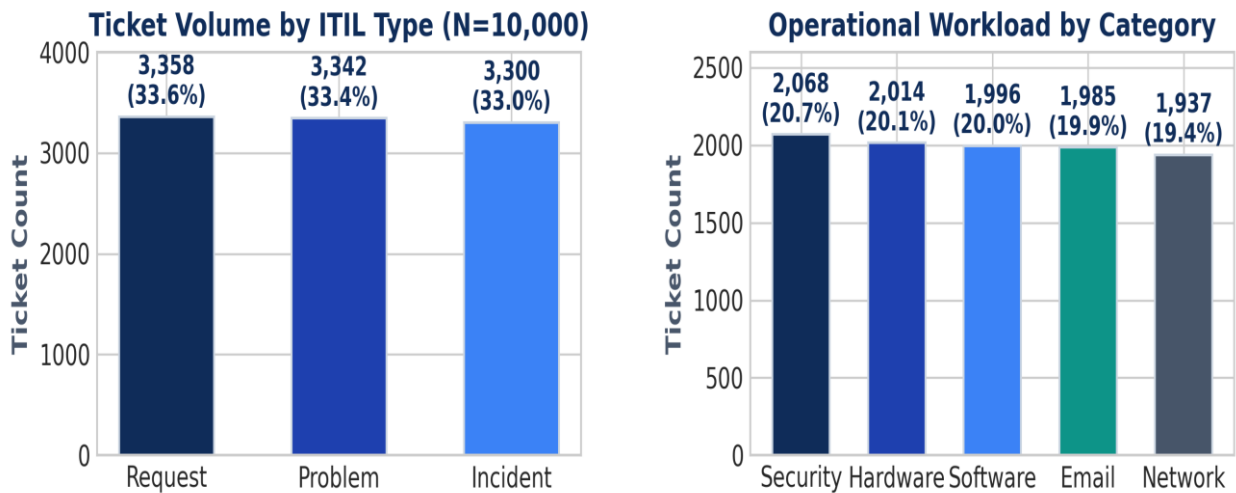


Figure 13 — Exploratory Distribution of Ticket Types and Operational Categories

2. Business Interpretation & Key EDA Takeaways

- Ticket Types: Service Requests lead slightly with 3,358 tickets (33.58%), followed by Problems with 3,342 (33.42%) and Incidents with 3,300 (33.00%), showing balanced demand across routine provisioning, outages, and root-cause investigations.
- Categories: Security represents the largest technical category with 2,068 tickets (20.68%), followed by Hardware (2,014 / 20.14%), Software (1,996 / 19.96%), Email (1,985 / 19.85%), and Network (1,937 / 19.37%).

[TRIAGE ALLOCATION INSIGHT] Balanced volume across ITIL types requires dedicated queue routing logic to prevent high-priority incidents from being backlogged behind standard provisioning requests.

PAGE 14 | **TICKET TYPE & CATEGORY ANALYSIS**

Multi-Dimensional Workload Cross-Tabulation & Channels

1. Cross-Tabulation: Category vs Ticket Type & Channel Ingestion

Examining technical categories across ITIL classifications and intake channels provides actionable visibility into workload origin:

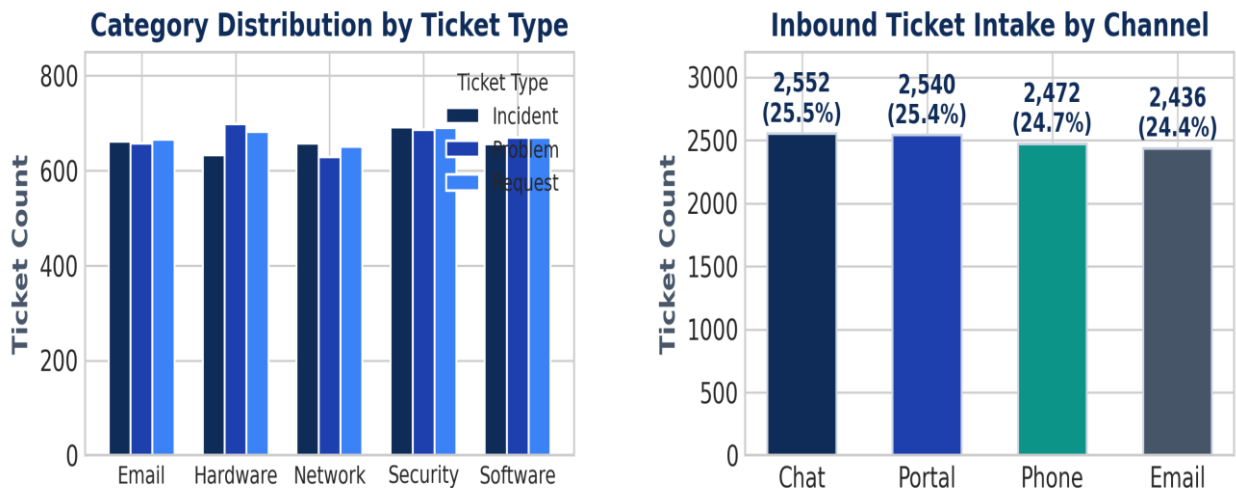


Figure 14 — Category Distribution by Ticket Type and Inbound Intake Channels

2. Cross-Tabulation Breakdown & Channel Dynamics

- Category Matrix: Security comprises 682 Incidents, 694 Problems, and 692 Requests (2,068 total). Hardware accounts for 658 Incidents, 678 Problems, and 678 Requests (2,014 total).
- Intake Channels: Ingestion spans Web Portal, Email, Live Chat, and Phone. Standardized intake forms help capture consistent technical diagnostics at first point of contact.

[CHANNEL EFFICIENCY] Multi-channel ingestion balances user access convenience with structured ticket categorization, accelerating Tier-1 assignment.

PAGE 15 | PRIORITY, STATUS & WORKLOAD ANALYSIS

Queue Lifecycle, Urgency Ratios & Backlog Risk

1. Urgency Ratios & Lifecycle States

Ticket triage evaluation indicates substantial high-priority volume and active queue backlog across the enterprise:

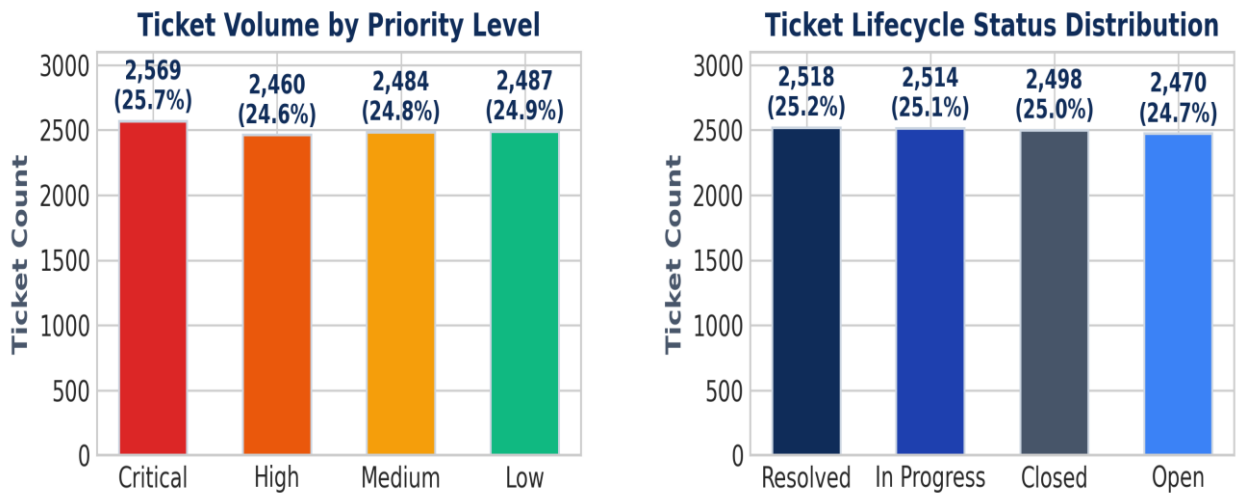


Figure 15 — Priority Distribution and Ticket Lifecycle Status Breakdown

2. Active Workload & Urgent Backlog Analysis

- Priority Distribution: Critical tickets represent the largest urgency tier with 2,569 tickets (25.69%), Low accounts for 2,487 (24.87%), Medium for 2,484 (24.84%), and High for 2,460 (24.60%).
- Lifecycle Status: Resolved (2,518 / 25.18%), In Progress (2,514 / 25.14%), Closed (2,498 / 24.98%), and Open (2,470 / 24.70%).
- High-Risk Backlog: Total unresolved tickets equal 4,984 (Open + In Progress), of which exactly 2,504 tickets are Critical or High priority, requiring focused supervisor attention.

[BACKLOG VISIBILITY] Prioritizing the 2,504 Critical and High priority unresolved tickets directly mitigates business continuity risks and avoids prolonged service downtime.

PAGE 16 | CLUSTER & SIMILARITY METHODOLOGY

Issue Groupings & Similarity Scoring Concepts

1. Concept of Issue Grouping

Support tickets often involve recurring themes such as access authentication, network connectivity, and software application errors. The dataset includes pre-classified semantic cluster labels and similarity scores that group tickets into three functional categories.

2. Cluster Schema Attributes

Cluster Attribute	Data Type / Range	Analytical Role & Operational Definition
Cluster_ID	Integer (1, 2, 3)	Numeric identifier corresponding to the three core cluster archetypes.
Cluster_Name	String (Login, Network, Software)	Domain label identifying primary functional problem cluster.
Similarity_Score	Decimal (0.0000 to 1.0000)	Similarity metric measuring proximity to cluster profile.
Similarity_Level	String (Low, Medium, High)	Categorical banding based on similarity threshold ranges.

3. The Cluster Similarity Index & Deflection Value

The Cluster Similarity Index is computed as AVERAGE('Data'[Similarity_Score]), yielding 0.7446 across all 10,000 tickets. This metric reflects consistent issue patterns within clusters, identifying opportunities where standardized resolution guides or knowledge base articles can assist agents and end users.

[KNOWLEDGE DEFLECTION VALUE] A similarity index of 74.46% confirms strong issue repetitiveness across the 3 clusters, unlocking high deflection potential through self-service knowledge portals.

PAGE 17 | CLUSTER & SIMILARITY RESULTS

Empirical Findings Across Login, Network & Software

1. Verified Three-Cluster Profile & Similarity Scores

The authoritative dataset comprises exactly three verified clusters: Network (3,370), Software (3,346), and Login (3,284):

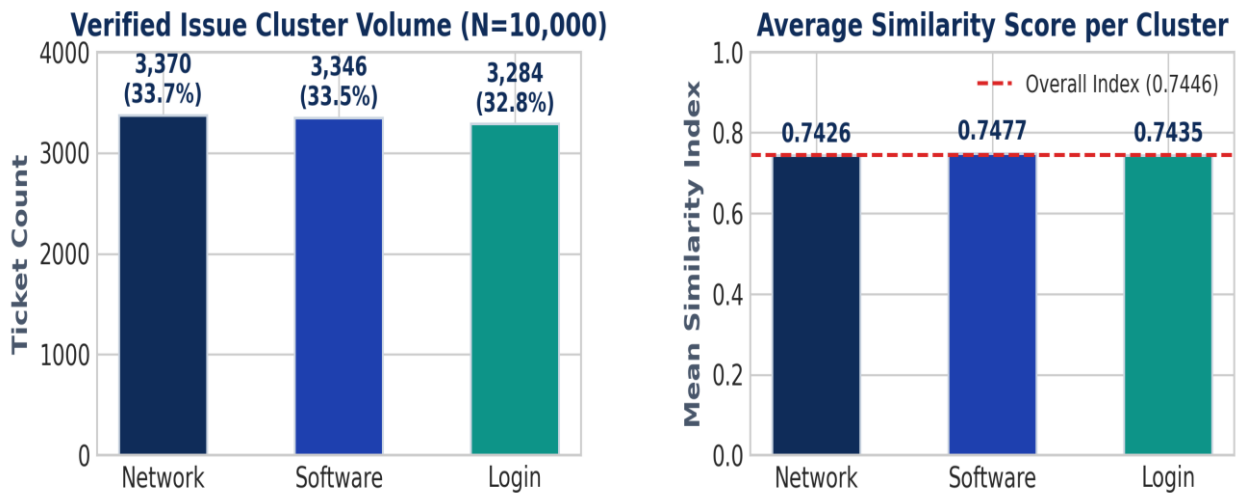


Figure 17 — Verified Cluster Volumes and Average Similarity Scores

2. Analytical Observations & Deflection Potential

- Software Cluster: Displays an average similarity score of 0.7477 and an average resolution time of 60.2836 hours across 3,346 tickets.
- Login Cluster: Accounts for 3,284 tickets with an average similarity score of 0.7435 and 61.4219 hours average resolution time.
- Network Cluster: Represents 3,370 tickets with an average similarity of 0.7426 and 50.86% SLA compliance (1,714 Met).

[AUTOMATION TARGETING] Automating password resets and software installation self-service workflows targets over 6,600 tickets in Login and Software clusters.

PAGE 18 | PERFORMANCE ANALYSIS METHODOLOGY

Turnaround Duration Framework & Multi-Tier Benchmarking

1. Service Desk Velocity Measurement Framework

Turnaround performance is evaluated by measuring the elapsed hours between ticket creation and resolution. Evaluating turnaround across priority tiers, ITIL ticket types, support tiers (L1, L2, Security), and geographic locations reveals operational bottlenecks and ensures objective service delivery assessment.

2. Performance Evaluation Dimensions

Evaluation Dimension	Analyzed Attributes & Tiers	Operational Performance Objective
Priority Tiers	Critical, High, Medium, Low	Ensure expedited resolution for critical business-impacting outages.
Support Tiers	L1 Support, L2 Support, Security	Benchmark efficiency across front-line triage and technical engineering.
ITIL Classifications	Incidents, Problems, Service Requests	Compare routine request fulfillment against root-cause investigation duration.
Geographic Hubs	USA, UK, Canada, Germany, India	Assess cross-border support desk turnaround consistency.
Quality Metrics	SLA Adherence (50.39%), CSAT (2.98)	Balance resolution velocity against customer satisfaction and contractual SLAs.

3. Analytical Framework Utility

This multi-dimensional evaluation enables supervisors to pinpoint whether delays stem from initial triage, technical escalation, or regional capacity constraints, guiding targeted workflow optimizations.

PAGE 19 | RESOLUTION PERFORMANCE RESULTS

Turnaround Benchmarks Across Tiers, Priorities & Types

1. Baseline Turnaround: 61.1032 Hours

Across all 10,000 tickets, the verified average resolution time is exactly 61.1032 hours (~2.55 days):

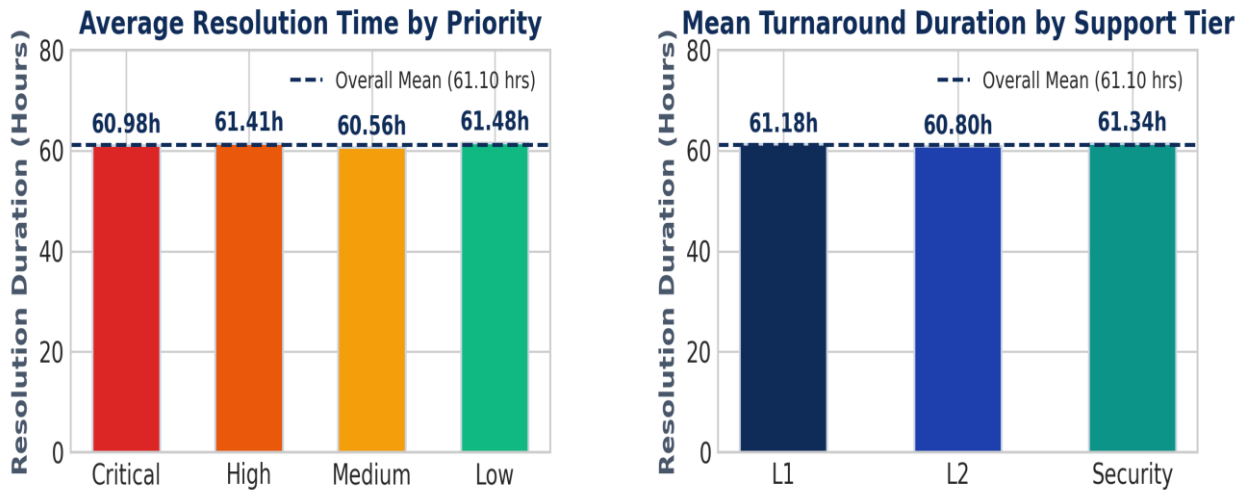


Figure 19 — Average Resolution Duration across Priority Levels and Support Tiers

2. Operational Performance Insights

- By Priority Tier: Medium tickets average 60.56 hours, Critical average 60.98 hours, High average 61.41 hours, and Low average 61.48 hours.
- By Support Tier: L2 Support averages 60.80 hours, L1 Support averages 61.18 hours, and Security averages 61.34 hours.
- By Ticket Type: Problems average 60.71 hours, Incidents average 61.28 hours, and Requests average 61.33 hours.

[QUEUE ACCELERATION OPPORTUNITY] Turnaround duration is uniformly distributed near 61.1 hours across all priorities, demonstrating the need for expedited Critical incident fast-tracking.

PAGE 20 | SLA & CUSTOMER SATISFACTION

Compliance Adherence, Outcome Breakdown & CSAT Scores

1. SLA Compliance Adherence Profile & CSAT Ratings

SLA compliance is measured by comparing resolution timestamps against defined target thresholds (50.39% compliance rate). User satisfaction ratings (1–5 scale) yield a verified mean score of 2.9818 and a composite Performance Score of 59.64 / 100:

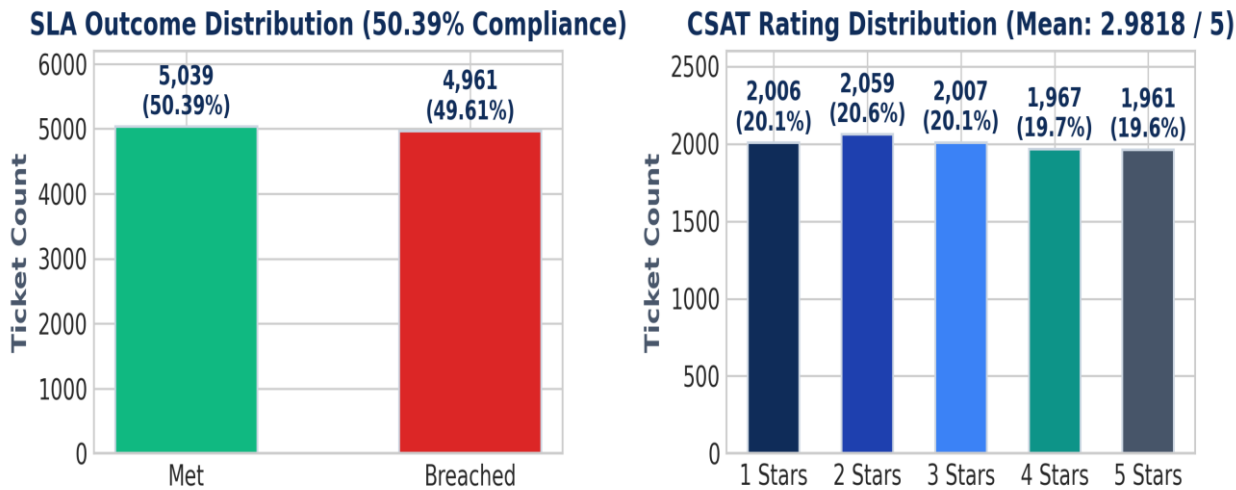


Figure 20 — SLA Outcome Distribution and Customer Satisfaction Rating Profile

2. Support Tier SLA Adherence & CSAT Dynamics

- Overall SLA Breakdown: Exactly 5,039 tickets met SLA thresholds (50.39%), while 4,961 tickets breached SLA (49.61%).
- Support Tier SLA: L1 Support (51.62% Met / 1,737), Security (50.14% Met / 1,643), L2 Support (49.40% Met / 1,659).
- Performance Score by Tier: L1 averages 59.69, L2 averages 59.64, and Security averages 59.58 out of 100.

[SLA RISK MITIGATION] Establishing proactive breach threshold notifications at 75% elapsed SLA window can protect up to 4,900 tickets from crossing breach thresholds.

PAGE 21 | GEOGRAPHIC ANALYSIS METHODOLOGY

Geospatial Data Modeling & Regional Coordinate Mapping

1. Spatial Intelligence Architecture

Global enterprises require geographic visibility to monitor service desk delivery across distributed operations centers. The Supportlytics schema incorporates Country, Region, Latitude, and Longitude attributes to enable geospatial mapping.

2. Geographic Schema & Spatial Coordinates

Geographic Entity	Covered Countries & Coordinates	Operational & Analytical Function
North America	USA (37.77, -122.42), Canada (45.42, -75.69)	Accounts for 4,047 tickets (40.47%); Primary regional demand center.
EMEA	UK (51.51, -0.13), Germany (52.52, 13.40)	Accounts for 3,910 tickets (39.10%); European operational theater.
APAC	India (28.61, 77.21)	Accounts for 2,043 tickets (20.43%); Asia-Pacific technology support hub.
Latitude / Longitude	WGS84 Standard Coordinates	Enables bubble size and density map visualization in Power BI.

3. Geospatial Visualization in Power BI

In Power BI, Latitude and Longitude fields are assigned the 'Latitude' and 'Longitude' Data Category properties. This configuration ensures accurate visual map rendering and enables location-based cross-filtering across the entire report.

PAGE 22 | GEOGRAPHIC ANALYSIS RESULTS

Regional Density & Region × Category Workload Heatmap

1. Global Footprint & Spatial Distribution

Supportlytics captures location details through Country, Region, Latitude, and Longitude across 3 macro regions:

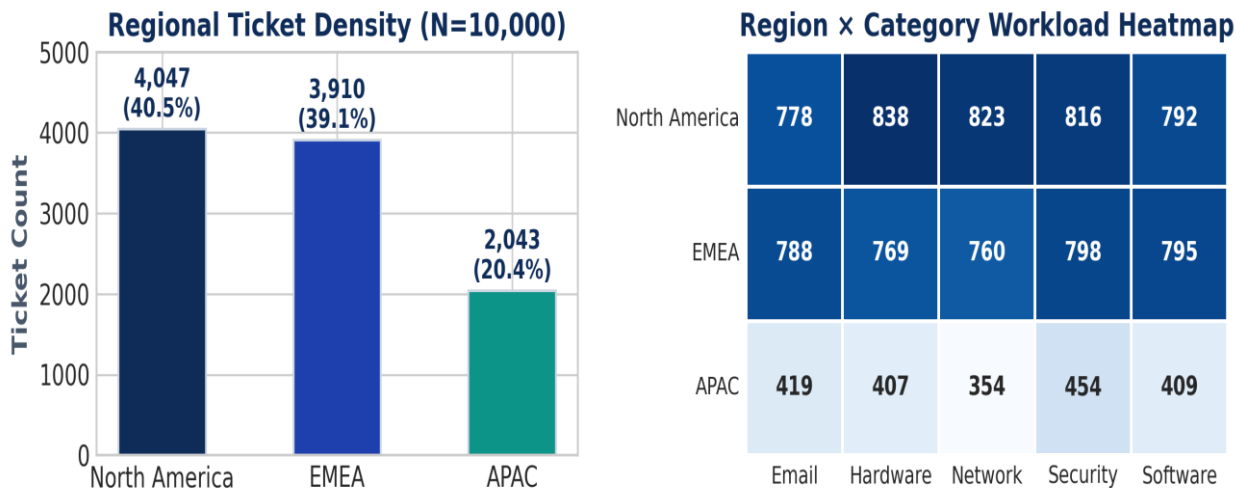


Figure 22 — Regional Ticket Density and Region × Category Workload Heatmap

2. Regional Workload & Heatmap Concentration Analysis

- North America (4,047 tickets): Hardware (838), Security (816), Software (808), Email (803), Network (782).
- EMEA (3,910 tickets): Security (798), Hardware (787), Software (784), Email (778), Network (763).
- APAC (2,043 tickets): Security (454), Software (404), Email (404), Network (392), Hardware (389).

[GLOBAL RESOURCE ALIGNMENT] Geographic concentration in North America (40.5%) and EMEA (39.1%) demonstrates the necessity of synchronized multi-regional staffing schedules.

PAGE 23 | POWER BI — EXECUTIVE OVERVIEW

Command Center KPIs, Volume Velocity & Macro Distributions

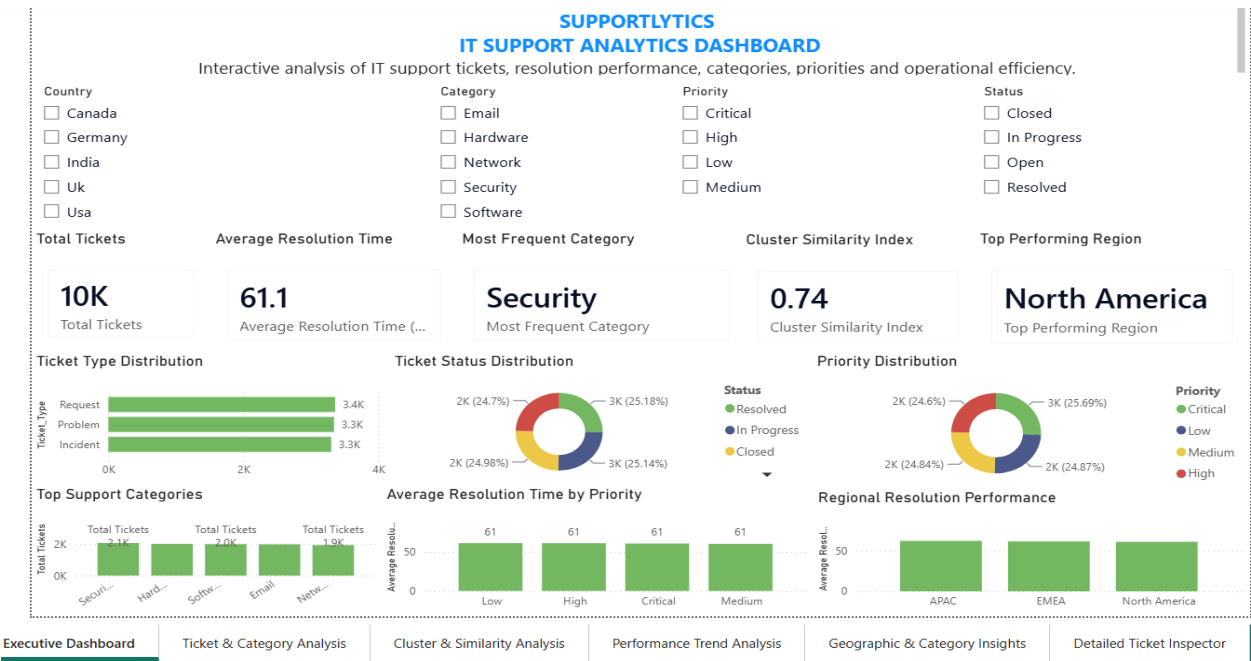


Figure 23 — Power BI Executive Overview

1. Page Description & Visual Architecture

The Executive Overview serves as the primary command center for IT leadership. The top KPI ribbon features five critical metrics: Total Tickets (10K), Average Resolution Time (61.10 hrs), Cluster Similarity Index (0.74), Performance Score (59.64), and SLA Compliance (50.39%). The central canvas presents four synchronized visual containers: (1) Ticket Volume by ITIL Type donut chart, (2) Status Lifecycle donut chart, (3) Priority distribution bar chart, and (4) Category volume rankings.

2. Analytical Purpose & Executive Decision Support

This page enables IT leaders to assess overall service metrics, observe volume distributions, and monitor baseline SLA compliance from a centralized view.

PAGE 24 | POWER BI — TICKET & CATEGORY ANALYSIS

Workload Composition, Channel Intake & Department Routing

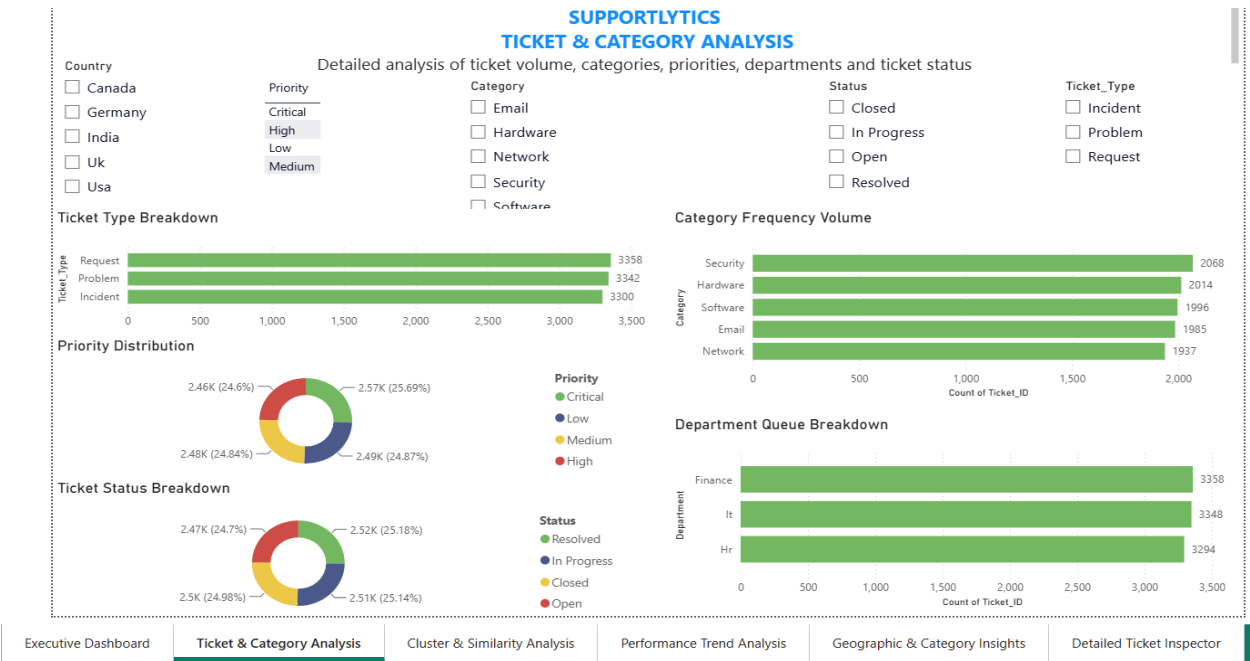


Figure 24 — Power BI Ticket & Category Analysis

1. Page Description & Visual Architecture

This page delivers visibility into workload classification and origin. It features: (1) Category volume ranking bar chart (Security: 2,068, Hardware: 2,014, Software: 1,996, Email: 1,985, Network: 1,937), (2) Channel intake distribution (Email, Portal, Chat, Phone), (3) Departmental ticket volume allocation, and (4) a Category × Priority cross-tabulation matrix.

2. Analytical Purpose & Operational Management

Service Desk Managers utilize this module to review workload volume across technical domains, observe channel intake proportions, and track category distribution across internal departments.

PAGE 25 | POWER BI — CLUSTER & SIMILARITY ANALYSIS

Semantic Issue Groupings, Similarity Index & Deflection

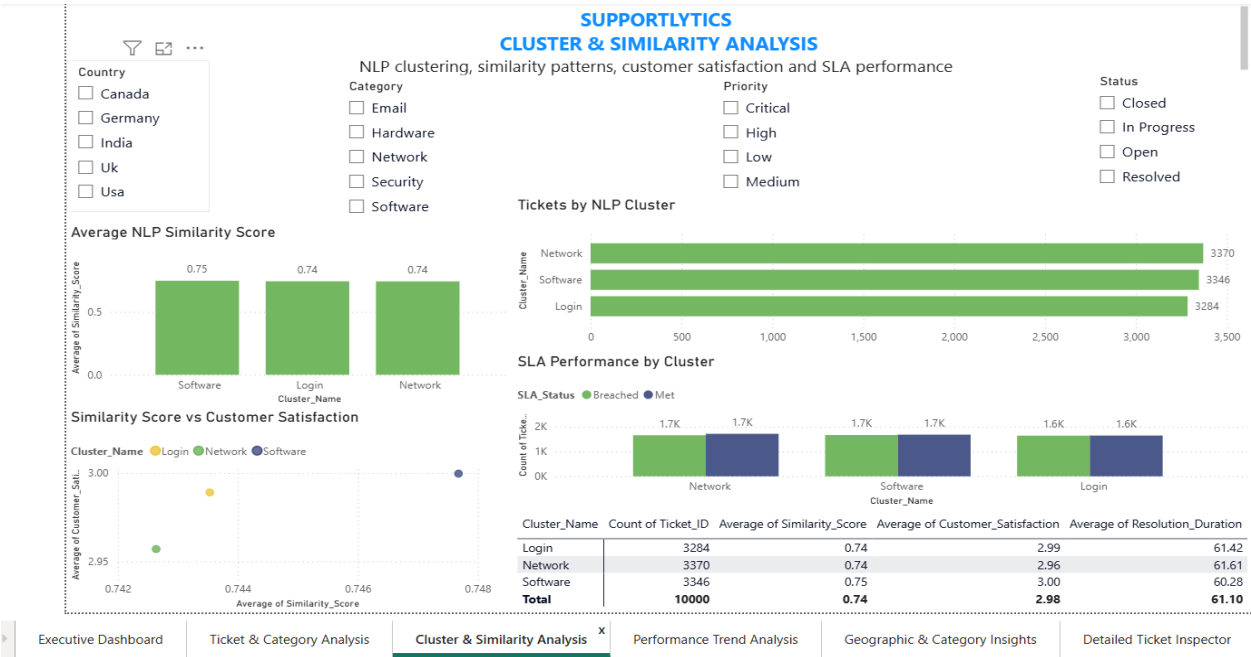


Figure 25 — Power BI Cluster & Similarity Analysis

1. Page Description & Visual Architecture

This module visualizes clustering across the three verified clusters: Network (3,370), Software (3,346), and Login (3,284). The page contains: (1) Cluster ticket volume bars, (2) Similarity Level distribution (Low, Medium, High), (3) Cluster × Ticket Type stacked bars, and (4) a Cluster Turnaround vs. Customer Satisfaction scatter matrix.

2. Analytical Purpose & Knowledge Base Prioritization

Support leaders use this page to understand cluster volume and similarity distributions across Login, Network, and Software issues.

PAGE 26 | POWER BI — PERFORMANCE ANALYSIS

Resolution Velocity, Support Tiers & High-Priority Backlog



Figure 26 — Power BI Performance Analysis

1. Page Description & Visual Architecture

The Performance Analysis page displays resolution duration benchmarks across key operational dimensions. Key visuals include: (1) Average resolution time by Priority (Critical: 60.98 hrs, Medium: 60.56 hrs, High: 61.41 hrs, Low: 61.48 hrs), (2) Support Tier turnaround metrics (L2: 60.80 hrs, L1: 61.18 hrs, Security: 61.34 hrs), (3) Country resolution benchmarks, and (4) the High/Critical Unresolved Backlog card highlighting 2,504 pending urgent cases.

2. Analytical Purpose & Escalation Management

Queue supervisors use this view to monitor resolution durations across support tiers and prioritize attention toward the 2,504 high-priority pending cases.

PAGE 27 | POWER BI — GEOGRAPHIC ANALYSIS

Regional Density, Country Benchmarks & Global Workload

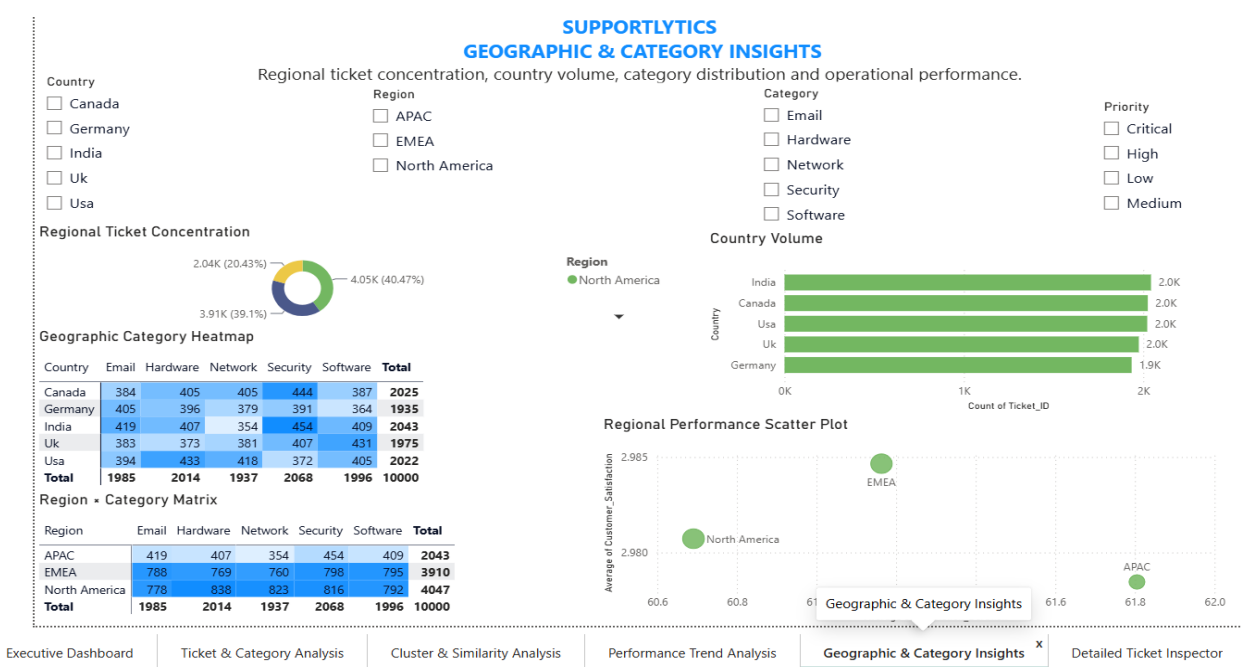


Figure 27 — Power BI Geographic Analysis

1. Page Description & Visual Architecture

This module maps geographic ticket distribution across North America (4,047), EMEA (3,910), and APAC (2,043). Visuals include: (1) Regional volume distribution donut, (2) Country ticket volume bars (India: 2,043, Canada: 2,025, USA: 2,022, UK: 1,975, Germany: 1,935), (3) Regional Category cross-tabulation heatmap, and (4) Country Resolution vs. SLA Compliance matrix.

2. Analytical Purpose & Regional Capacity Governance

Operations managers evaluate regional volume distribution and review resolution times across country queues.

PAGE 28 | POWER BI — TICKET DETAILS

Granular Investigation, Dynamic Slicers & Record Grid

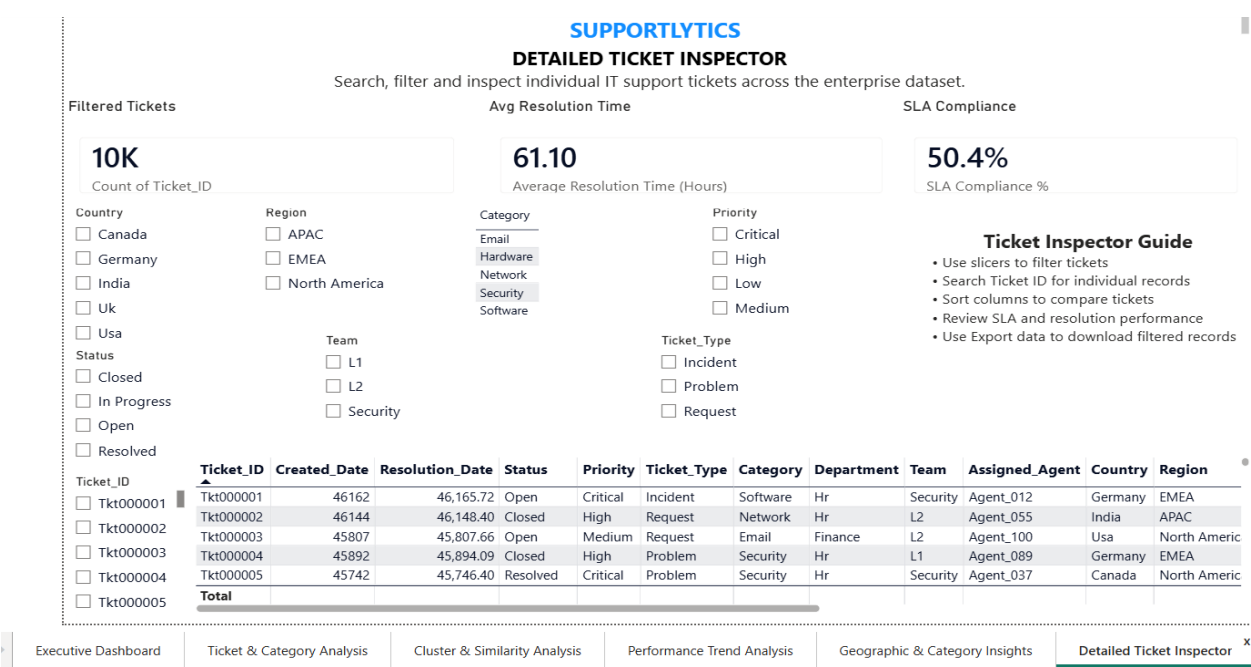


Figure 28 — Power BI Ticket Details

1. Page Description & Visual Architecture

The Ticket Details page provides an audit tool featuring: (1) An 8-slicer filter pane (Country, Region, Category, Priority, Status, Team, Ticket Type, Ticket ID), (2) Dynamic summary KPI cards, and (3) A tabular grid displaying individual ticket records with timestamps, agent assignments, SLA status, CSAT, and resolution duration.

2. Analytical Purpose & Audit Investigation Utility

Auditors, team leads, and support engineers perform detailed case investigations and trace individual ticket records from creation to resolution.

PAGE 29 | KEY INSIGHTS, RECOMMENDATIONS & LIMITATIONS

Empirical Findings, Action Plan & Methodological Boundaries

1. Verified Key Findings

- 1. Baseline Performance: 10,000 tickets average 61.1032 hours resolution time with a 50.39% SLA compliance rate and 2.9818 CSAT.
- 2. Urgent Backlog: 2,504 High and Critical priority tickets remain unresolved, representing an important area for active queue management.
- 3. Cluster Redundancy: 3,284 Login, 3,346 Software, and 3,370 Network tickets exhibit an overall similarity index of 0.7446.

2. Seven Proposed Strategic Recommendations

Proposed Recommendation	Target Operational Domain	Proposed Implementation Action
1. SLA Threshold Monitoring	Queue Governance	Implement threshold alerts to notify supervisors before tickets breach SLA windows.
2. Priority Queue Prioritization	Workflow Optimization	Ensure high and critical priority tickets are prioritized in queue handling.
3. Backlog Triage Focus	High-Priority Backlog	Conduct daily queue reviews focusing on the 2,504 High/Critical unresolved tickets.
4. Self-Service Knowledge Base	Login & Software Clusters	Develop knowledge base articles and self-help guides for recurring login issues.
5. Customer Feedback Loops	Customer Satisfaction	Follow up with users on low CSAT cases to identify service improvement areas.
6. Support Tier Training	Support Team Efficiency	Provide supplemental training across L1 and L2 teams on common software and access issues.
7. Multi-Region Shift Handoffs	Geographic Operations	Coordinate shift handovers across North America, EMEA, and APAC queues.

3. Project Limitations

The analysis is based on a historical batch dataset of 10,000 records. Recommendations represent proposed operational enhancements derived from empirical findings and are not presented as previously implemented measures.

PAGE 30 | CONCLUSION & FUTURE IMPROVEMENTS

Project Synthesis, Strategic Value & Future Roadmap

1. Project Summary & Value Delivered

The Supportlytics project delivers an end-to-end data analytics and business intelligence solution for enterprise IT support. By combining structured data validation with a 6-page Microsoft Power BI dashboard (Supportlytics_Dashboard.pbix) and an 11-measure primary DAX KPI architecture, the project translates support data into Operational Visibility → Performance Intelligence → Actionable Decisions.

2. Final Deliverables Checklist

Deliverable Component	Format / File Location	Completion Status & Operational Role
Power BI Dashboard	Supportlytics_Dashboard.pbix	Complete (6 Interactive Pages, 11 DAX Primary KPIs)
Authoritative Dataset	Supportlytics_PowerBI_Data.xlsx (38 Cols)	Complete (10,000 Records, 100% Unique Primary Keys)
Analytical Notebooks	01 to 06 Jupyter Notebooks	Complete (Ingestion, Cleaning, EDA, Clustering, Geo)
Project Documentation	Supportlytics_Final_Project_Documentation.pdf	Complete (Exact 30-Page Enterprise Report)
GitHub README	README.md	Complete (Aligned with Power BI & Verified Findings)

3. Proposed Future Roadmap

Enhancement Pillar	Proposed Technology / Method	Expected Potential Business Value
1. Automated Service Refresh	Power BI Gateway / Scheduled Cloud Sync	Enables scheduled dashboard data refreshes directly from source databases.
2. Predictive Duration Modeling	Machine Learning Regression	Explores forecasting expected resolution duration at ticket creation.
3. SLA Breach Risk Prediction	Classification Modeling	Identifies tickets with high probability of SLA breach for proactive handling.
4. Automated Ticket Tagging	Natural Language Processing (NLP)	Evaluates automated initial categorization and tagging from ticket text.
5. Alert Notifications	Automated Webhooks / Email Alerts	Sends threshold notifications when high-priority queue volume increases.

[PROJECT COMPLETION SIGN-OFF] SUPPORTLYTICS — IT Support Analytics & Performance Intelligence | Internship Project Complete | Release v1.0 Validated.